

Oishik Sengupta

Senior Backend Engineer | Node.js · TypeScript · JavaScript · AWS · Distributed Systems
Pune, India | oishik8sengupta@gmail.com | +91 6291758741 | linkedin.com/in/oishiksengupta

SUMMARY

Senior Backend Engineer with 4 years of experience in the travel industry, building distributed, event-driven systems for OTA and travel-tech platforms using Node.js, TypeScript, JavaScript, AWS and React. Leads backend delivery for the Flights LOB, mentoring a team of five engineers.

SKILLS

Languages & Frameworks: Node.js, TypeScript, JavaScript, React.js, Jest

Databases & Caching: DynamoDB, Redis Enterprise, PostgreSQL, MySQL

AWS: Lambda, SQS, SNS, EventBridge, API Gateway, EC2, S3, CloudWatch, ECS, ECR

Architecture & Practices: Event-driven architecture, microservices, distributed systems, HLD/LLD, SOLID principles, REST/SOAP APIs

Tools: Git, Jenkins, GitHub Actions, CI/CD, JIRA, Elasticsearch, Kibana, Grafana, Zenduty

EXPERIENCE

Senior Software Developer — udChalo

Jun 2025 – Present

- Own backend delivery for the Flights team across supplier integrations, core business logic and production support using Node.js, TypeScript, AWS and React.
- Designed and shipped an automated pending-booking reconciliation system on SNS, SQS and Lambda, cutting pending booking recovery from 30–45 minutes of manual work per booking to seconds.
- Independently designed a Redis-backed supplier session pool with lifecycle management, cutting supplier login volume from one per search to low double digits per carrier per day.
- Built a configuration-driven pricing system using DynamoDB and the Strategy pattern, with a React admin interface for fees, markups and insurance rules, contributing to a 3% revenue increase.

Senior Software Engineer — AnalytixKraft

Mar 2024 – May 2025

- Independently designed and built an event-driven flight status notification service using AWS Lambda, DynamoDB, RDS, SNS and webhooks, supporting ~3,000 bookings per day and 100–200 actively subscribed flights at peak.
- Added automated monitoring and alerts using Elasticsearch, Kibana, Grafana and Zenduty across booking and payment workflows, reducing incident detection time by 30% and improving SLA resolution times.
- Led the migration to Redis Enterprise from ElastiCache, cutting infrastructure costs by over ₹1 lakh a year without affecting reliability.

Software Developer → Senior Software Developer — AlgoDomain Solutions

Aug 2022 – Feb 2024

- Promoted to Senior Software Developer for owning backend work across booking, payment and airline integration.
- Wrote PostgreSQL scripts to create table views using stored procedures, then added TypeORM code to the existing TypeScript reporting service to execute those procedures and generate reports.

PROJECTS

Automated Pending-Booking Reconciliation — AWS Lambda, SNS, SQS, API Gateway, Node.js, TypeScript, React

- Solved bookings stranded in pending when airline responses exceeded the API Gateway 29-second timeout, leaving a valid PNR at the supplier with no database record.
- Published every PNR to SNS so reconciliation runs independently for all bookings, removing failure detection from the critical request path.
- Built a reconciliation flow that compares supplier and database booking status, confirms successful bookings, and keeps pending bookings in SQS until confirmation or retry timeout.
- Used Server-Sent Events (SSE) to update the React confirmation page in real time, moving bookings from pending to confirmed or refunded without manual refresh.

Distributed Supplier Session Pooling & Lifecycle Management — Redis Cluster, AWS Lambda, Node.js, TypeScript

- Built a Redis-backed session pool with atomic checkout and MULTI/EXEC transitions, ensuring one session is assigned to only one request at a time.
- Used per-supplier Redis hash tags to keep related keys in the same Redis Cluster slot, allowing multi-key transactions to work correctly.
- Built a safe return flow that clears and verifies session state before reuse, with FIFO pooling and session reservation to prevent concurrent sharing.

Flight Status Notification Service — AWS Lambda, DynamoDB, RDS, SNS, API Gateway, WhatsApp Meta API

- Deduplicated provider subscriptions to one per flight with internal fan-out to passengers, reducing duplicate subscriptions and provider cost.
- Used polyglot persistence with DynamoDB for flight status and subscription lookups, and RDS for user-to-flight subscription mappings.
- Secured supplier webhooks with token authentication and optimistic locking, with retries and timestamp-based handling for duplicate or out-of-order events.

EDUCATION

Bachelor of Technology, Electronics & Communication Engineering — Future Institute of Engineering and Management | 2018 – 2022
CGPA: 8.45